13. **Ans: C**

Average wavelength of light = 550 nm.

Using Rayleigh criterion, limiting angle $= \dfrac{550 \times 10^{-9}}{2.8 \times 10^{-3}} = 1.964 \times 10^{-4}$ rad

Using small angle approximation, taking the distance between pixels as the "arc length",

$s = r\theta$

$$r = \frac{s}{\theta} = \frac{3.0 \times 10^{-4}}{1.964 \times 10^{-4}} = 1.5 \text{ m}$$

Note if wavelength of light is not given:
Using wavelength as 400 nm, distance = 2.1 m
Using wavelength as 700 nm, distance = 1.2 m
Hence answer must lie between 1.2 m to 2.1 m.

14 Ans: A